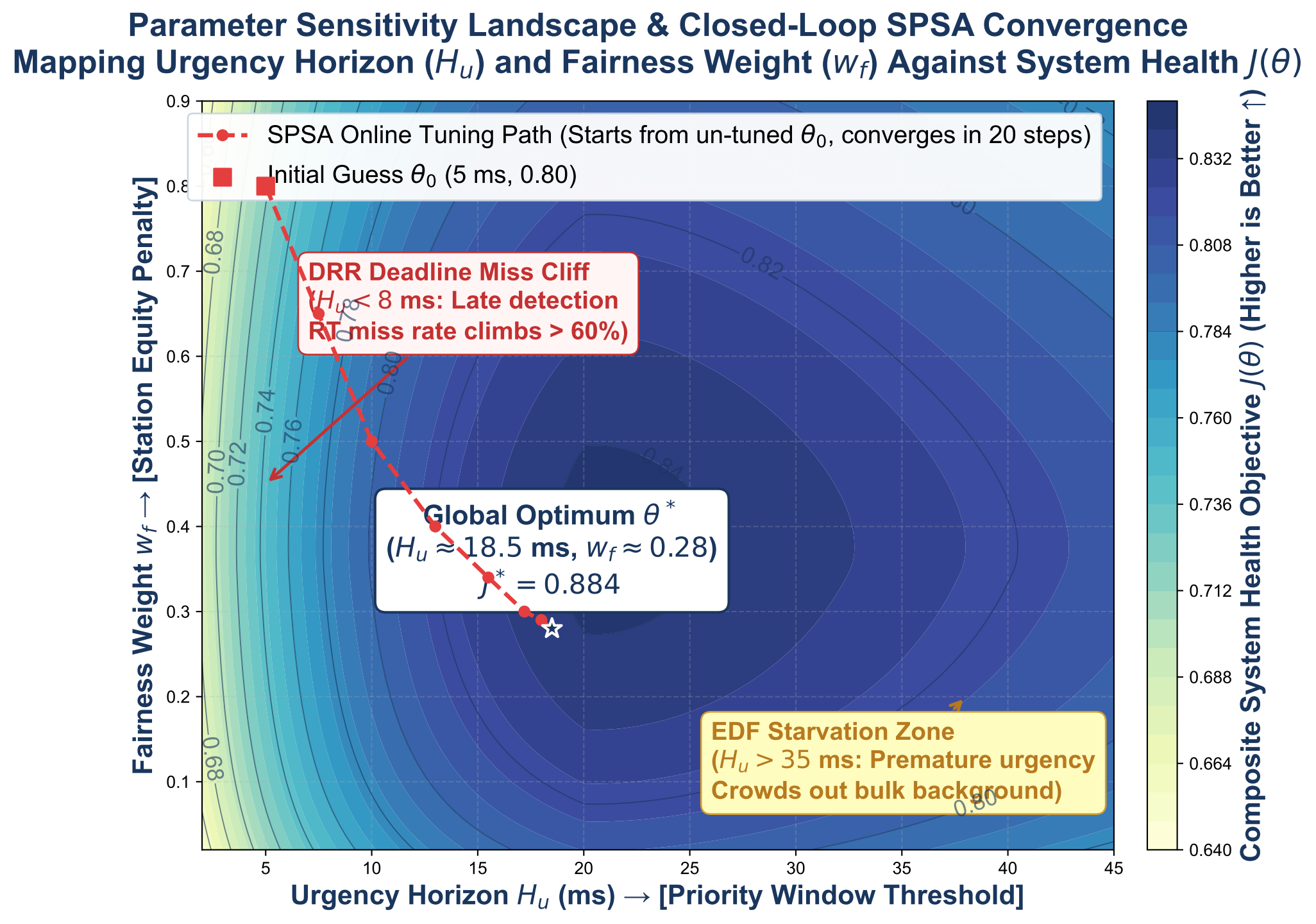

Parameter Sensitivity Landscape & Closed-Loop SPSA Convergence
Mapping Urgency Horizon (H_u) and Fairness Weight (w_f) Against System Health $J(\theta)$
SPSA Online Tuning Path (Starts from un-tuned θ_0, converges in 20 steps)
Initial Guess θ_0 (5 ms, 0.80)
EDRR Deadline Miss Cliff
($H_u < 8$ ms: Late detection
RT miss rate climbs > 60%)
Global Optimum θ^*
($H_u \approx 18.5$ ms, $w_f \approx 0.28$)
$J^* = 0.884$
EDF Starvation Zone
($H_u > 35$ ms: Premature urgency
Crowds out bulk background)
Fairness Weight $w_f \rightarrow$ [Station Equity Penalty]
Urgency Horizon H_u (ms) $\rightarrow$ [Priority Window Threshold]
Composite System Health Objective $J(\theta)$ (Higher is Better ↑)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
5
10
15
20
25
30
35
40
45
0.832
0.808
0.784
0.760
0.736
0.712
0.688
0.664
0.640
0.68
0.70
0.72
0.74
0.76
0.78
0.80
0.82
0.84
0.86
0.88